

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456392.77819 +/- 0.00053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1583 +/- 0.0085
Transit depth (Rp/Rs)^2	2.51 +/- 0.27 [%]
Orbital Inclination (inc)	87.12 +/- 0.86
Airmass coefficient 1 (a1)	0.9187 +/- 0.0096
Airmass coefficient 2 (a2)	0.0414 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	#1 - [422, 381]
Optimal Aperture	3.31
Optimal Annulus	8.13
Transit Duration (day)	0.06046 +/- 0.00093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1583 $\pm$ 0.0085 vs 0.1594 (deviation 0.13 σ)
Duration fit vs published	0.06046 d vs 0.0512 d (deviation 9.6 σ)
Mid-time O-C	-0.0 min
Depth SNR	18.0
Residual scatter	1.06 %

Light curve

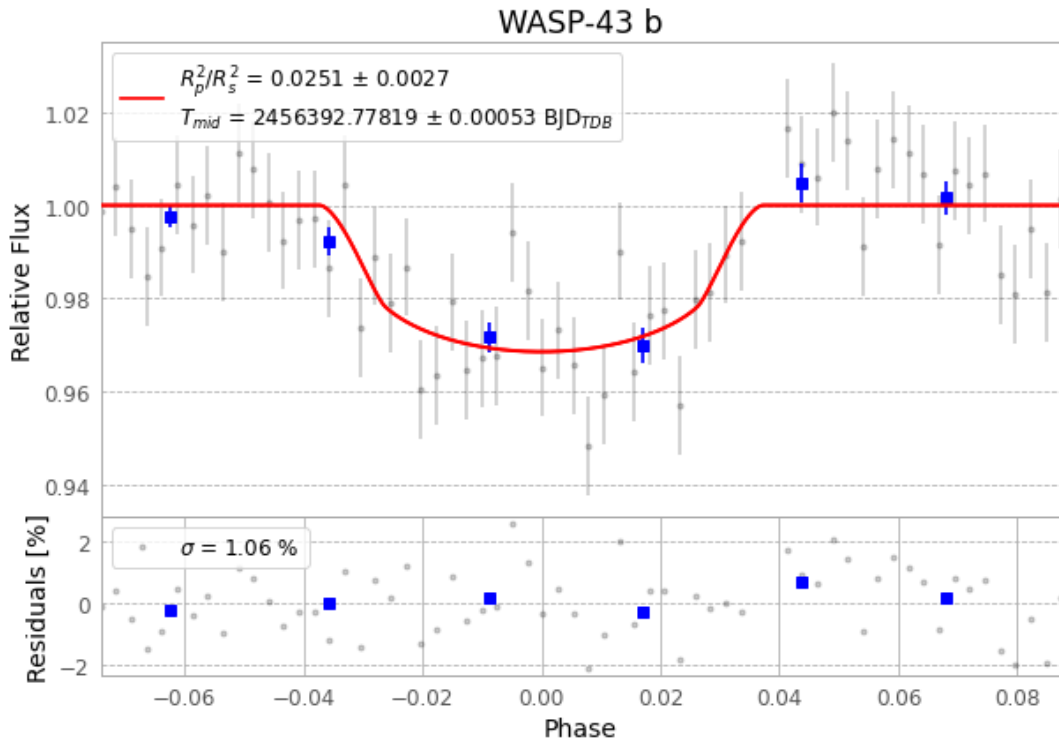


Figure 1 — FinalLightCurve_WASP-43 b_2013-04-09.png (SHA-256 3e2c4d348fce21c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [336, 335], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6157554d3269eea9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43130410050613.FITS → WASP-43130410081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130410.log (5d3b751b9284...), FinalLightCurve_WASP-43 b_2013-04-09.png (3e2c4d348fce...), FinalLightCurve_WASP-43 b_2013-04-09.csv (77e8443d9072...)

Compute resources

Wall time 154.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 23:33Z.